AUCKLAND INSTITUTE OF STUDIES NEW ZEALAND

COURSE OUTLINE

SOFT703 WEB APPLICATIONS DEVELOPMENT

PREREQUISITES: None
NZQF Level: 7
Credits: 15

LECTURER:	HAVEA FONUA
Room:	ROOM A118
Phone:	815-1717 ext. 851
Email:	haveaf@ais.ac.nz
Contact hour:	Tuesday 12:30 p.m. – 1:30 p.m. ALSO, OUTSIDE OF THIS TIME BY APPOINTMENT

COURSE AIM

The aim of this course is to extend the students' knowledge of data communications and internetworking over the World Wide Web, to cover issues associated with development of applications in this environment. Emphasis is upon the analysis, design and development of web-based applications for a variety of purposes using current tools and techniques. The multi-tiers system architecture including presentation layer, web services layer, business logic layer, data access layer and database layer will be discussed.

LEARNING OUTCOMES

The learners will be able to:

1. Perform a proper analysis and design of a given business case study for implementation in Microsoft .NET Technology.
2. Design and develop the presentation layer for the web application using: ASP.Net, HTML5, CSS and Java-script.
3. Implement JQuery and AJAX in the presentation layer using ASP.Net technology.
4. Develop data access layer (DAL) and Business logic layer (BLL) and then construct web applications with web service component to access either business logic layer (BLL) or Data access layer (DAL) of given business case study.
5. Configure & deploy web applications on a web-server. Configure IIS Server.
6. Develop workplace soft-skills including working in groups, writing formal reports, carrying out individual research and/or delivering oral presentations.

COURSE DURATION

The course will be conducted over a 14-week period with a typical teaching week consisting of 4 hours of classes.

Classes will be held at the following times:

Tuesday	10:00 am – 12:00 pm	R102
Wednesday	10:00 am – 12:00 pm	R102

You are required to have **100% attendance**, in accordance with programme requirements.

LEARNING HOURS

The course involves 150 hours of learning: 48 hours of structured in-class learning and 102 of associated learning activities such as reading course materials, preparing for class, working on assignments, or peer group discussions and reviewing feedback or material. An indicative time allocation is shown below.

Activity	**Hours**
In class instruction	48
Reading course materials	25
Working on assignments	52
Preparation for assessments	25
TOTAL	150

COURSE CONTENT

Week # Starting	**Session**	**Topics**
Week 1 15 September	1 2	• Course introduction • Requirements Analysis, Design & Modeling • Introduction to ASP.NET web programming
Week 2 22 September	3 4	• Three Tiers Architecture, MVC with HTML 5 and CSS3 • Testing and debugging ASP.NET
Week 3 15 September	5 6	• Applications Standard Server Controls, Javascript • Validation controls, JQUERY
Week 4 6 October	7 8	• State, Cookies and URL encoding • Master Page, MVC Master View, Razor View
Week 5 13 October	9 10	• Themes, Ajax • Site Navigation and ASP.NET Routing
Week 6 20 October	11 12	• Introduction to Database Programming, Entity Framework • View Component 1 - GridView Control
Week 7 27 October	13 14	• Revision for Midterm Test • Midterm Test
Week 8 3 November	15 16	• View Component 2 - DetailsView and FormView controls • View Component 3 - ListView and DataPager controls

Week 9 10 November	17 18	• Object Data Sources with ADO.NET • Securing a Website, Web Service
Week 10 17 November	19 20	• Authenticating and User Authorization • Use Email, Custom Error Pages, and Back-Button Control
Week 11 24 November	21 22	• Web Server with IIS and Cloud • Configure and Publish ASP.NET Applications
Week 12 1 December	23 24	• Revision
Week 13 8 December		**EXAM WEEK**
Week 14 15 December		**EXAM WEEK**

The dates following week numbers are for the Monday of that week.

The following are public holidays and there will be no classes or assessments on these days:

- Monday 27 October, Labour Day

Note: The above schedule is provided as a general guide and may be subject to minor variation.

READINGS AND RESOURCES

Your textbook will be issued in e-book format, not as a hard copy. You will receive instructions for using a code to download your e-book onto your device. **Please note that e-books, once downloaded, cannot be returned.** If there is any possibility that you might change your course, do not download the e-book; wait until you have made your final course selection decisions for the trimester before downloading any e-books. If you download the e-book for a course, and then withdraw from the course, you will be required to pay extra for the e-book for the new course.

Online Learning Management System

The AIS learning management system is Moodle which can be accessed at the following address: https://moodle.ais.ac.nz/

Moodle is one of the ways AIS optimises your learning experience. It is a platform where you can manage your learning. Please take the time to familiarise yourself with each section of Moodle so that you feel comfortable navigating yourself around. Teaching materials and assessments are all normally distributed via Moodle. You should access Moodle before, and routinely throughout, your course.

Prescribed texts:
None

Recommended reading:
Delamater, M. and Murach, J. (2022). Murach's ASP.NET Core MVC, 2nd Edition, Mike Murach & Associates, ISBN 978-1-943873-02-9

Delamater, M. and Boehm, A. (2016). *Murach's ASP.NET 4.6 Web Programming with C# 2015*, 6th Edition, Mike Murach & Associates, ISBN: 13: 978-1890774950

Ragupathi, M. T. (2016). *Learning ASP. NET Core MVC Programming*. Packt Publishing Ltd.

Resources
Students should also refer to other resources some of which are listed below:
https://msdn.microsoft.com/

COURSE ASSESSMENT

The assessment for this course will be as follows:

Title	Course Marks %	Release Date	Due date	Learning Outcomes	Corresponding Course Content
Assignment 1	30 %	19/09/2025 Week2	24/10/2025 Week6	1, 2, 4, 6	CLASS 1-14
Midterm Test	30 %	29/10/2025 Week7		2, 4	CLASS 1-12
Assignment 2	40 %	31/10/2025 Week7	05/12/2025 Week12	2, 3, 4, 5, 6	CLASS 1-24
Total	**100 %**				

Submission of written assignments are made through the submission boxes in Moodle.

You have the option of submitting written assessments as draft submissions to check for similarity matches prior to making final submissions. The draft submission option closes 24 hours prior to the due date and cannot be extended. You will not be able to submit your assessment unless you have attended APA Referencing Skills and Avoiding Plagiarism workshops and passed the Good Referencing Test (GRT).

Passing Requirements

- You must gain at least 40% in the assignment 2 assessment, and an overall mark of 50% or higher, to pass this course **and**;
- You must achieve all learning outcomes.

LATE SUBMISSION PENALTIES

The due date(s) are as stated above. Any assignment submitted late – after 4.00pm on the due date for submission, or approved extension date – is subject to a 5% deduction of the available marks from the marks awarded for each successive 24-hour period (or part period) that has elapsed and

will receive a mark of zero if submitted after 96 hours have elapsed. If you have concerns about your ability to meet a deadline, you are advised to consult with your lecturer before the due date.

See the guidelines for your programme in the AIS Student Handbook for more details should you require an assignment extension.

LEARNING SUPPORT

Studying at this level requires awareness and understanding of academic requirements which can be, at times, difficult to comprehend. Several support options are available:

1. Consult with your course lecturer, either in person, via e-mail, or via Moodle
2. Contact the Learning Hub for appropriate support (learninghub@ais.ac.nz)
3. Contact the library (ais-library@ais.ac.nz)
4. Contact the Head of School, Information Technology
5. Contact your class representative

Please refer to the AIS Student Handbook for other support services that are available. The handbook can be found on the AIS website (www.ais.ac.nz).

All materials within the workbook and on Moodle are copied under the AIS Copyright Licence and Print Media Licence. Copied material has been copied solely for the educational purpose of AIS. Please use the materials for your own use in this course.

Recorded lessons will be posted on Moodle for your personal use only and are not for redistribution.

It is not acceptable for students to record lessons without the full knowledge of all the people in the classroom. Please ask permission before recording classes. Provided it is reasonable, permission will normally be given.

We recommend that you retain your course outlines for future reference as they are subject to modification.